Homework 1

1. **Conda test data**

```
lina@gen:~$ conda create -n nf bioconda::nextflow
2 channel Terms of Service accepted
Retrieving notices: done
Channels:
 - conda-forge
 - bioconda
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done


==> WARNING: A newer version of conda exists. <==
    current version: 26.1.0
    latest version: 26.3.1

Please update conda by running

    $ conda update -n base -c conda-forge conda



## Package Plan ##

  environment location: /home/users/lina/miniconda3/envs/nf

  added / updated specs:
    - bioconda::nextflow
```

```
Proceed ([y]/n)? y


Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate nf
#
# To deactivate an active environment, use
#
#     $ conda deactivate

lina@gen:~$ conda activate nf
(nf) lina@gen:~$
```

```
(nf) lina@gen:~$ nextflow run nextflow-io/rnaseq-nf -profile conda

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [big_laplace] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/users/lina/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/users/lina/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[4e/d52e3a] RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[f1/51a92c] RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[2f/c10e75] RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[f1/d5e59d] MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 19:38:00
Duration    : 1m 11s
CPU hours   : (a few seconds)
Succeeded   : 4
```

2. Docker test data

```
(nf) lina@gen:/mnt/Storage/Wolf/metaspades_new/welcome-to-docker$ nextflow run nextflow-io/rnaseq-nf -profile docker --outdir ./results_docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [reverent_mercator] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/users/lina/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/users/lina/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : ./results_docker

executor >  local (4)
[21/0d5d6a] RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[5f/848f15] RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[7b/45ae03] RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[a1/e9a574] MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> ./results_docker/multiqc_report.html

Completed at: 09-Apr-2026 20:02:37
Duration    : 1m 45s
CPU hours   : 0.1
Succeeded   : 4
```

3. Custom data (SRR390728)

```
(nf) lina@gen:/mnt/Storage/Wolf/metaspades_new/welcome-to-docker$ nextflow run nextflow-io/rnaseq-nf     --reads 'SRR390728_{1,2}.fastq'     --trans
criptome /mnt/Storage/Wolf/metaspades_new/welcome-to-docker/Homo_sapiens.GRCh38.cdna.all.fa     --outdir ./results_human     -profile docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [desperate_lumiere] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /mnt/Storage/Wolf/metaspades_new/welcome-to-docker/Homo_sapiens.GRCh38.cdna.all.fa
reads        : SRR390728_{1,2}.fastq
outdir       : ./results_human

executor >  local (4)
[a5/bbf501] RNASEQ:INDEX (Homo_sapiens) [100%] 1 of 1 ✔
[db/08a214] RNASEQ:FASTQC (SRR390728)   [100%] 1 of 1 ✔
[37/38530f] RNASEQ:QUANT (SRR390728)    [100%] 1 of 1 ✔
[05/0c7600] MULTIQC                     [100%] 1 of 1 ✔

Done! Open the following report in your browser --> ./results_human/multiqc_report.html

Completed at: 09-Apr-2026 20:51:41
Duration    : 7m 48s
CPU hours   : 0.1
Succeeded   : 4
```

Done ;)